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## CAN-open protocol

A Controller Area Network (CAN bus) is a vehicle bus standard designed to allow microcontrollers and devices to communicate with each other in applications without a host computer. It is a message-based protocol, designed originally for multiplex electrical wiring within automobiles, but is also used in many other contexts.

CANopen is a communication protocol and device profile specification for embedded systems used in automation. In terms of the OSI model, CANopen implements the layers above and including the network layer. The CANopen standard consists of an addressing scheme, several small communication protocols and an application layer defined by a device profile. The communication protocols have support for network management, device monitoring and communication between nodes, including a simple transport layer for message segmentation/desegmentation. The lower level protocol implementing the data link and physical layers is usually Controller Area Network (CAN), although devices using some other means of communication (such as Ethernet Powerlink, EtherCAT) can also implement the CANopen device profile.

### Use and benefits of CANopen

CANopen protocol allows multiple controllers to be connected into an extensible unified

network. Its flexible configuration capabilities offer easy access to exposed device parameters and real-time automatic (cyclic or event-driven) data transfer.

The benefits of CANopen include:

- Standardized in EN50325-4
- Widely supported and vendor independent
- Highly extensible
- Offers flexible structure (can be used in a wide variety of application areas)
- Suitable for decentralized architectures
- Wide support of CANopen monitoring tools and solutions

### Common Configurations

**CAN Mode:** Used to select one of the 3 operating modes. Off disables all CAN receive and transmit capabilities.

**Node ID:** CAN Node ID used for transmission from the controller. Value may be between 1 and 126 included.

**Bit Rate:** Selectable bit rate. Available speeds are 1000, 800, 500, 250, 125, 50, 25, 10 kbit/s. Default is 125kbit and is the recommended speed for RawCAN and MiniCAN modes.

**Heartbeat:** Period at which a Heartbeat frame is sent by the controller. The



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frame is CANopen compatible 0x700 + NodeID, with one data byte of value 0x05 (Status: Operational). The Heartbeat is sent in any of the selected modes. It can be disabled by entering a value of 0.

### Supported CAN Modes

Three CAN operating modes are available on the CAN-enabled controllers:

- 1 - RawCAN
- 2 - MiniCAN
- 3 - CANopen

**RawCAN** is a low-level operating mode giving read and write access to CAN frames. It is recommended for use in low data rate systems that do not obey to any specific standard.

CAN frames are typically built and decoded using the MicroBasic scripting language.

**MiniCAN** is greatly simplified subset of CANopen, allowing, within limits, the integration of the controller into an existing CANopen network. This mode requires MicroBasic scripting to prepare and use the CAN data.

CANopen is the full Standard from CAN in Automation (CIA), based on the DS302 specification. It is the mode to use if full compliance with the CANopen standard is a primary requisite.

This section describes the RawCAN and MiniCAN modes, refer to section “CANopen Interface” for a description of the CANopen mode.

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### CAN Connection

Connection to a CAN bus is as simple as shown on the diagram above. 120 Ohm Termination Resistors must be inserted at both ends of the bus cable. CAN network can be up to 1000m long. See CAN specifications for maximum length at the various bit rates.



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## CAN Connection

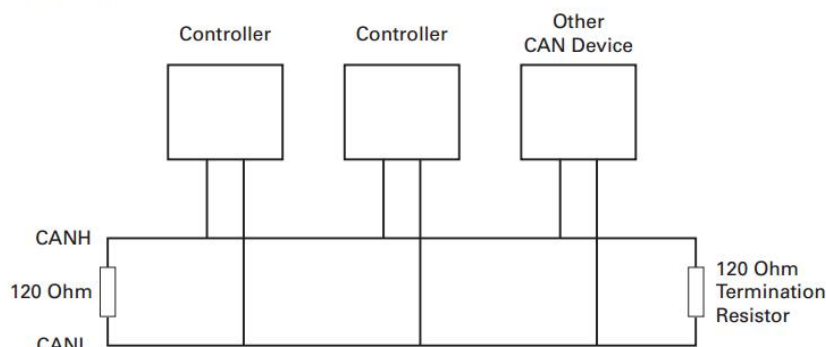


FIGURE 60. CAN connection

## CAN Bus Configuration

Though KeyaMotorMonitor software, The controller configures the communication parameters Must be though RS232 to PC.

Use the CAN menu in the Configuration tab in order to enable the CANopen mode. Additionally, the utility can be used to configure the following parameters:

Default configurationL:

- Standard frame
- CAN type: CANopen
- Node ID: 1
- Bit rate:250 kbps
- Heartbeat (ms):1000ms
- Autostart:Enable
- TPDO Enable and Send rate:1500ms

## CAN Bus Pinout

Depending on the controller model, the CAN signals are located on the 15-pin female connector or 9-pin male connector. Refer to datasheet for details.

### CAN Signals on DB15 connector

| Pin Number | Signal | Description  |
|------------|--------|--------------|
| 6          | CAN_L  | CAN bus low  |
| 7          | CAN_H  | CAN bue high |

### CAN Signals on DB25 connector

| Pin Number | Signal | Description  |
|------------|--------|--------------|
| 23         | CAN_H  | CAN bus HIGH |
| 24         | CAN_L  | CAN bus LOW  |

## CAN and USB Limitations

On most controller models CAN and USB cannot operate at the same time. On



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controllers equipped with a USB connector, if simultaneous connection is not allowed, the controller will enable CAN if USB is not connected.

The controller will automatically enable USB and disable CAN as soon as the USB is connected to the PC. The CAN connection will then remain disabled until the controller is restarted with the USB unplugged.

See the controller model datasheet to verify whether simultaneous CAN and USB is supported

The controller operating in the CANopen mode can accept the following types of messages:

- Service Data Objects, or SDO messages to read/write parameter values
- Process Data Objects, or PDO mapped messages to automatically transmit parameters and/or accept commands at runtime
- Network Management, or NMT as defined in the CANopen specification

### **Service Data Object (SDO) Read/Write Messages**

Runtime queries and runtime commands can be sent to the controller in real-time using the expedited SDO messages.

SDO messages provide generic access to Object Dictionary and can be used for obtaining parameter values on an irregular basis due to the excessive network traffic that is generated with each SDO request and response message.

### **Transmit Process Data Object (TPDO) Messages**

Transmit PDO (TPDO) messages are one of the two types of PDO messages that are used during operation.

TPDOs are runtime operating parameters that are sent automatically on a periodic basis from the controller to one or multiple nodes. TPDOs do not alter object data; they only read internal controller values and transmit them to the CAN bus.

### **Receive Process Data Object (RPDO) Messages**

RPDOs are configured to capture runtime data destined to the controller  
CANopen implementation supports RPDOs. Data received using RPDOs are stored in 8 user variables from where they can be processed using MicroBasic scripting.

### **CANopen data transmitting and receiving**

Use CANopen to send and receive data , is not a simple specified ID to send commands and parameters, need to meet certain load specification.

#### **Read node Date**

#### **The frame**

CANopen compatible 0x700 + NodeID,(0x01-0x7e) Range :0x601-0x67E

| Data_0 | Data_1、 2 | Data_3    | Data_4 | Data_5 | Data_6 | Data_7 |
|--------|-----------|-----------|--------|--------|--------|--------|
| 0x40   | index     | sub-index | 00     | 00     | 00     | 00     |

The parameters of the target node returns according to the frame format



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| Data_0 | Data_1、2 | Data_3    | Data_4 | Data_5 | Data_6 | Data_7 |
|--------|----------|-----------|--------|--------|--------|--------|
| 0x4F   | index    | sub-index | XX     | 00     | 00     | 00     |
| 0x4B   | index    | sub-index | XX     | XX     | 00     | 00     |
| 0x47   | index    | sub-index | XX     | XX     | XX     | 00     |
| 0x43   | index    | sub-index | XX     | XX     | XX     | XX     |

Read Node parameter

Loading ways of the ID: 0 x600 + target address ID (0 x01-0 x7e), range of 0x601-0x67E

Loading ways of the data frame is as follows:

| Data_0 | Data_1、2 | Data_3    | Data_4 | Data_5 | Data_6 | Data_7 |
|--------|----------|-----------|--------|--------|--------|--------|
| 0x2F   | index    | sub-index | XX     | 00     | 00     | 00     |
| 0x2B   | index    | sub-index | XX     | XX     | 00     | 00     |
| 0x27   | index    | sub-index | XX     | XX     | XX     | 00     |
| 0x23   | index    | sub-index | XX     | XX     | XX     | XX     |

Note: 0 x2F means message sending a byte of data effectively, and so on, 0 x23 means four bytes of data packets sent.

The target node returned by the CAN data frame format

| Data_0 | Data_1、2 | Data_3    | Data_4 | Data_5 | Data_6 | Data_7 |
|--------|----------|-----------|--------|--------|--------|--------|
| 0x60   | index    | sub-index | 00     | 00     | 00     | 00     |

Note: This script does not check for loss of communication on the CAN bus. It is provided for information only



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Note:1. The "XX" said one byte of data effectively."00" said invalid bytes, the data is not standard.

2. The CANopen sending and receiving data fields are eight bytes, extra byte is invalid.

3. Multi-byte index, data are low byte in the front.

4. Send invalid command will return an error frame, knowledge is not in the scope of this agreement is introduced.

### (3) command example

#### 1. Speed open control

Explain: Instruction index for 0x2000, Speed value range(-1000---1000), to 1000 is forward max speed, to 0 is output is 0, to -1000 is reverse max speed。

If you want to control with ID 1 controller, the speed of the open-loop mode, need to send the message as below

ID : 0x601

Data:

Speed value of 500 (0x01F4)

|      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|
| 0x23 | 0x00 | 0x20 | 0x01 | 0xf4 | 0x01 | 0x00 | 0x00 |
|------|------|------|------|------|------|------|------|

Speed value of 1000 (0x03E8)

|      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|
| 0x23 | 0x00 | 0x20 | 0x01 | 0xE8 | 0x03 | 0x00 | 0x00 |
|------|------|------|------|------|------|------|------|

Speed value of -500 (0x0FE0C, 即0x01F4的补码+1)

|      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|
| 0x23 | 0x00 | 0x20 | 0x01 | 0x0C | 0xFE | 0x00 | 0x00 |
|------|------|------|------|------|------|------|------|

Speed value of -1000 (0x0FC18, 即0x03E8的补码+1)

|      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|
| 0x23 | 0x00 | 0x20 | 0x01 | 0x18 | 0xFC | 0x00 | 0x00 |
|------|------|------|------|------|------|------|------|

Reply message as below

ID : 0x581

Data :

|      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|
| 0x60 | 0x00 | 0x20 | 0x01 | 0x00 | 0x00 | 0x00 | 0x00 |
|------|------|------|------|------|------|------|------|



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2、Set position mode target value 4000000 (0x3D0900)

|      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|
| 0x23 | 0x01 | 0x20 | 0x01 | 0x00 | 0x09 | 0x3D | 0x00 |
|------|------|------|------|------|------|------|------|

3、Set position mode speed 1000 (0x03E8)

|      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|
| 0x23 | 0x02 | 0x20 | 0x01 | 0xE8 | 0x03 | 0x00 | 0x00 |
|------|------|------|------|------|------|------|------|

4、Read position counter

|      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|
| 0x40 | 0x04 | 0x21 | 0x01 | 0x00 | 0x00 | 0x00 | 0x00 |
|------|------|------|------|------|------|------|------|

5、Read run speed

|      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|
| 0x40 | 0x03 | 0x21 | 0x01 | 0x00 | 0x00 | 0x00 | 0x00 |
|------|------|------|------|------|------|------|------|

6、stop commend

|      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|
| 0x2C | 0x0E | 0x20 | 0x01 | 0x01 | 0x00 | 0x00 | 0x00 |
|------|------|------|------|------|------|------|------|

When Test, it is recommended to use the CAN card to send packets and monitor the bus status.

## Object Dictionary

| index  | sub | Entry Name             | Data Type & Access | Command Reference |
|--------|-----|------------------------|--------------------|-------------------|
| 0x2000 | 01  | Set motor commend,ch 1 | S16 WO             | “G”               |
|        | 02  | Set motor commend,ch 2 |                    |                   |
| 0x200C | 00  | Emergency Shutdown     | U8 WO              | “EX”              |
| 0x200D | 00  | Release Shutdown       | U8 WO              | “MG”              |
| 0x2C0E | 01  | Set motor commend,ch 1 | U8 WO              | “MS”              |
|        | 02  | Set motor commend,ch 2 |                    |                   |
| 0x2017 | 00  | Save Config to Flash   | U8 WO              | “EES”             |

### Instruction index form

| index  | sub | Entry Name                 | Data Type & Access | Command Reference |
|--------|-----|----------------------------|--------------------|-------------------|
| 0x2001 | 01  | Set Position commend, ch.1 | S32 WO             | “p”               |
|        | 02  | Set Position commend, ch.2 |                    |                   |
| 0x2002 | 01  | Set Position speed, ch.1   | S16 WO             | “S”               |
|        | 02  | Set Position speed, ch.2   |                    |                   |
| 0x2003 | 01  | Set Encoder Counter, ch.1  | S32 WO             | “C”               |
|        | 02  | Set Encoder Counter, ch.2  |                    |                   |
| 0x2006 | 01  | Set Acceleration , ch.1    | S32 WO             | “MAC”             |
|        | 02  | Set Acceleration , ch.2    |                    |                   |



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|        |    |                         |        |        |
|--------|----|-------------------------|--------|--------|
| 0x2007 | 01 | Set Deceleration , ch.1 | S32 WO | “MDEC” |
|        | 02 | Set Deceleration , ch.2 |        |        |

Instruction index form

| index  | sub | Entry Name                                          | Data Type & Access | Command Reference |
|--------|-----|-----------------------------------------------------|--------------------|-------------------|
| 0x2100 | 01  | Read Motor Amps, ch.1                               | S16 RO             | “A”               |
|        | 02  | Read Motor Amps, ch.2                               | S16 RO             |                   |
| 0x2101 | 01  | Read Actual Motor Command, ch.1                     | S16 RO             | “M”               |
|        | 02  | Read Actual Motor Command, ch.2                     |                    |                   |
| 0x2103 | 01  | Read Encoder Motor Speed, ch.1                      | S16 RO             | “S”               |
|        | 02  | Read Encoder Motor Speed, ch.2                      | S16 RO             |                   |
| 0x2107 | 01  | Read Encoder Motor Speed as 1/1000 of Max, ch.1     | S16 RO             | “SR”              |
|        | 02  | Read Encoder Motor Speed as 1/1000 of Max, ch.2     |                    |                   |
| 0x2108 | 01  | Read Encoder Count Relative, ch.1                   | S32 RO             | “CR”              |
|        | 02  | Read Encoder Count Relative, ch.2                   |                    |                   |
| 0x2109 | 01  | Read Brushless Count Relative, ch.1                 | S32 RO             | “CBR”             |
|        | 02  | Read Brushless Count Relative, ch.2                 |                    |                   |
| 0x210A | 01  | Read BL Motor Speed in RPM, ch.1                    | S16 RO             | “BS”              |
|        | 02  | Read BL Motor Speed in RPM, ch.2                    |                    |                   |
| 0x210C | 01  | Read Battery Amps, ch.1                             | S16 RO             | “BA”              |
|        | 02  | Read Battery Amps, ch.2                             |                    |                   |
| 0x210D | 01  | Read Internal Voltages (V Int)                      | U16 RO             | “V”               |
|        | 02  | Read battery Voltages (v)                           | U16 RO             |                   |
|        | 03  | Read Internal Voltages ( mV)                        | U16 RO             |                   |
| 0x210E | 00  | Read All Digital Inputs                             | U32 RO             | “D”               |
| 0x210F | 01  | Read Case & Internal Temperatures (MCU Temperature) | S8 RO              | “T”               |
|        | 02  | Read Case & Internal Temperatures (ch.1)            |                    |                   |
|        | 03  | Read Case & Internal Temperatures (ch.2)            |                    |                   |
| 0x2111 | 00  | Read Status Flags                                   | U8 RO              | “FS”              |
| 0x2112 | 00  | Read Fault Flags                                    | U8 RO              | “FF”              |
| 0x2113 | 00  | Read Current Digital Outputs                        | U8 RO              | “DO”              |
| 0x2119 | 00  | Read time                                           | U32 RO             | “TM”              |
| 0x2121 | 00  | Read motor Status                                   | U8 RO              | “FM”              |





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**General agreement details:**

**EX - Emergency Stop**

CANOpen id: 0x200C

Description:

The EX command will cause the controller to enter an emergency stop in the same way as if hardware emergency stop was detected on an input pin. The emergency stop condition will remain until controller is reset or until the MG release command is received.

Syntax Serial: !EX

**MG - Emergency Stop Release**

CANOpen id: 0x200D

Description:

The MG command will release the emergency stop condition and allow the controller to return to normal operation. Always make sure that the fault condition has been cleared before sending this command

Syntax Serial: !MG

**MS - Stop in all modes**

CANOpen id: 0x200E

Description:

The MS command is similar to the EX emergency stop command except that it is applied to the specified motor channel

Syntax Serial: !MS [cc] Motor Channel

**EES - Save Configuration in EEPROM**

CANOpen id: 0x2017

Description:

This command causes any changes to the controller's configuration to be saved to Flash. Saved configurations are then loaded again next time the controller is powered on. This command is a duplication of the EESAV maintenance command. It is provided as a Real-Time command as well in order to make it possible to save configuration changes from within MicroBasic scripts.

Syntax Serial: !EES

**P - Go to Absolute Desired Position**

CANOpen id: 0x2001

Description:

This command is used in the Position Count mode to make the motor move to a specified encoder or hall count value. Runtime Commands

Syntax Serial: !P [cc] nn

cc = Motor channel

nn = Absolute count destination

Example:

!P 1 10000 : make motor go to absolute count value 10000.

**S - Set Motor Speed**



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CANOpen id: 0x2002

Description:

In the Closed-Loop Speed mode, this command will cause the motor to spin at the desired RPM speed. In Closed-Loop Position modes, this commands determines the speed at which the motor will move from one position to the next. It will not actually start the motion.

Syntax Serial: !S [cc] nn

Argument 1: Channel

Min: 1 Max: Total Number of Motors

Argument 2: Value Type: Signed 32-bit

Min: -500000 Max: 500000

### **C - Set Encoder Counters**

CANOpen id: 0x2003

Description:

This command loads the encoder counter for the selected motor channel with the value contained in the command argument. Beware that changing the controller value while operating in closed-loop mode can have adverse effects.

Syntax Serial: !C [cc] nn

cc = Motor channel

nn = Counter value

Example:

!C 2 -1000 : Loads -1000 in encoder counter 2

!C 1 0 : Clears encoder counter 1

### **AC - Set Acceleration**

Alias: ACCEL HexCode: 07 CANOpen id: 0x2006

Description:

Set the rate of speed change during acceleration for a motor channel. This command is identical to the MACC configuration command but is provided so that it can be changed rapidly during motor operation. Acceleration value is in  $0.1 * \text{RPM per second}$ . When using controllers fitted with encoder, the speed and acceleration value are actual RPMs. Brushless motor controllers use the hall sensor for measuring actual speed and acceleration will also be in actual RPM/s. When using the controller without speed sensor, the acceleration value is relative to the Max RPM configuration parameter, which itself is a user-provided number for the speed normally expected at full power. Assuming that the Max RPM parameter is set to 1000, and acceleration value of 10000 means that the motor will go from

0 to full speed in exactly 1 second, regardless of the actual motor speed.

Syntax Serial: !AC cc nn

cc = Motor channel

nn = Acceleration value in  $0.1 * \text{RPM/s}$

Example:

!AC 1 2000 : Increase Motor 1 speed by 200 RPM every second if speed is measured by encoder



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!AC 2 20000 : Time from 0 to full power is 0.5s if no speed sensors are present and Max RPM is set to 1000

### **DC - Set Deceleration**

CANOpen id: 0x2007

Description:

Set the rate of speed change during deceleration for a motor channel. This command is identical to the MDEC configuration command but is provided so that it can be changed rapidly during motor operation. Deceleration value is in  $0.1 * \text{RPM}$  per second. When using controllers fitted with encoder, the speed and deceleration value are actual RPMs.

Brushless motor controllers use the hall sensor for measuring actual speed and deceleration will also be in actual RPM/s. When using the controller without speed sensor, the deceleration value is relative to the Max RPM configuration parameter, which itself is user-provided number for the speed normally expected at full power. Assuming that the Max RPM parameter is set to 1000, and deceleration value of 10000 means that the motor will go from full speed to 0 in exactly 1 second, regardless of the actual motor speed.

Syntax Serial: !DC cc nn

cc = Motor channel

nn = Deceleration value in  $0.1 * \text{RPM/s}$

Example:

!DC 1 2000 : Reduce Motor 1 speed by 200 RPM every second if speed is measured by encoder

!DC 2 20000 : Time from full power to stop is 0.5s if no speed sensors are present and Max RPM is set to 1000

### **A - Read Motor Amps**

Alias: MOTAMPS HexCode: 00 CANOpen id: 0x2100

Description:

Measures and reports the motor Amps for all operating channels. Note that the current flowing through the motors is often higher than this flowing through the battery.

Syntax Serial: ?A [cc]

cc = Motor channel

aa = Amps \*10 for each channel

Example:

Q: ?A

R: A=100:200

Q: ?A 2

R: A=200

### **V - Read Volts**

CANOpen id: 0x210D

Description:

Reports the voltages measured inside the controller at three locations: the main



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battery voltage, the internal voltage at the motor driver stage, and the voltage that is available on the 5V output on the DSUB 15 or 25 front connector. For safe operation, the driver stage voltage must be above 12V. The 5V output will typically show the controller's internal regulated 5V minus the drop of a diode that is used for protection and will be in the 4.7V range. The battery voltage is monitored for detecting the undervoltage or overvoltage conditions.

Syntax Serial: ?V [ee]

ee =

1 : Internal volts

2 : Battery volts

3 : 5V output

nn = Volts \* 10 for internal and battery volts. Millivolts for 5V output

Example:

Q: ?V

R:V=135:246:4730

Q: ?V 3

R:V=4730

### **C - Read Encoder Counter Absolute**

CANOpen id: 0x2104

Description:

Returns the encoder value as an absolute number. The counter is 32-bit with a range of +/- 2147483648 counts.

Reply:

C=nn Type: Signed 32-bit Min: -2147M Max: 2147M

Where:

cc = Encoder channel number

nn = Absolute counter value

### **T - Read Temperature**

CANOpen id: 0x6403

Description:

Reports the temperature at each of the Heatsink sides and on the internal MCU silicon chip. The reported value is in degrees C with a one degree resolution.

Syntax Serial: ?T [cc]

Reply:

T= cc Type: Signed 8-bit Min: -40 Max: 125

Where:

cc =

1 : MCU temperature

2 : Channel 1 side

3 : Channel 2 side

tt = temperature in degrees

Note:

On some controller models, additional temperature values may reported. These



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are measured at different points and not documented. You may safely ignore this extra data. Other controller models only have one heatsink temperature sensor and therefore only report one value in addition to the Internal IC temperature.

### **FS - Read Status Flags**

CANOpen id: 0x6405

Description:

Report the state of status flags used by the controller to indicate a number of internal conditions during normal operation. The response to this query is the single number for all status flags. The status of individual flags is read by converting this number to binary and look at various bits of that number.

Syntax Serial: ?FS

$FS = f1 + f2*2 + f3*4 + \dots + fn*2^{n-1}$  Type: Unsigned 8-bit Min: 0 Max:

255

Where:

f1 = Serial mode

f2 = Pulse mode

f3 = Analog mode

f4 = Power stage off

f5 = Stall detected

f6 = At limit

f7 = Unused

f8 = MicroBasic script running

FF - Read Fault Flags

### **CANOpen id: 0x6406**

Description:

Reports the status of the controller fault conditions that can occur during operation. The response to that query is a single number which must be converted into binary in order to evaluate each of the individual status bits that compose it.

Syntax Serial: ?FF

$FF = f1 + f2*2 + f3*4 + \dots + fn*2^{n-1}$

Type: Unsigned 8-bit Min: 0 Max: 255

Where:

f1 = Overheat

f2 = Overvoltage

f3 = Undervoltage

f4 = Short circuit

f5 = Emergency stop

f6 = Brushless sensor fault

f7 = MOSFET failure

f8 = Default configuration loaded at startup

Example:

Q: ?FF

R: FF=2 : Overvoltage fault



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### **FM - Read Runtime Status Flag**

Alias: MOTFLAG HexCode: 30 CANOpen id: 0x2122

Description:

Report the runtime status of each motor. The response to that query is a single number which must be converted into binary in order to evaluate each of the individual status bits that compose it.

Syntax Serial: ?FM [cc]

cc = Motor channel

f1 = Amps Limit currently active

f2 = Motor stalled

f3 = Loop Error detected

f4 = Safety Stop active

f5 = Forward Limit triggered

f6 = Reverse Limit triggered

f7 = Amps Trigger activated

Example:

Q: ?FM 1

R: FM=6 : Motor 1 is stalled and loop error detected

Note:

On controller models supporting Spektrum radio mode f4 is used to indicate Spektrum. f4 to f6 are shifted to f5 to f7